

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pharm. Examination

University Theory Examination November 2017

PH317 PHARMACEUTICAL ANALYSIS-I

Date: 25.11.2017, Saturday

Time: 10 a.m. to 1.00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. 20

1. pH of the solution of salt derived from weak acid and strong base can be calculated using...
[A] $\text{pH} = \frac{1}{2} [\text{pK}_w - \text{pK}_b - \log [\text{NH}_4^+]]$
[B] $\text{pH} = \frac{1}{2} [\text{pK}_w - \text{pK}_a - \log [\text{Ac}^-]]$
[C] $\text{pH} = \frac{1}{2} [\text{pK}_w + \text{pK}_a - \text{pK}_b]$
[D] $\text{pH} = \frac{1}{2} [\text{pK}_w + \text{pK}_a + \log [\text{Ac}^-]]$
2. The preferred indicator in Fajan's titrimetric method is _____, which requires acidic medium.
[A] Fluorescein
[B] Ferric ammonium sulphate
[C] Potassium chromate
[D] Dichlorofluorescein
3. What is the composition of Karl Fischer reagent?
[A] 1 mole Iodine :3 moles sulphur dioxide :9 moles pyridine
[B] 1 mole Iodine :4 moles sulphur dioxide :12 moles pyridine
[C] 2 moles Iodine :3 moles sulphur dioxide :10 moles pyridine
[D] 1 mole Iodine :3 moles sulphur dioxide :10 moles pyridine
4. Reproducibility of methods can be expressed by _____.
[A] Selectivity
[B] Accuracy
[C] Precision
[D] Sensitivity
5. Which one is aprotic solvent used in non-aqueous titration?
[A] Water
[B] Glacial acetic acid
[C] Benzene
[D] Perchloric acid

6. In gravimetric method during a precipitation, crystal growth can be promoted by _____.
 [A] Precipitating from a cool solution
 [B] Precipitating from a basic solution
 [C] Precipitating from a dilute solution
 [D] Precipitating from a neutral solution
7. In a complexometric titration, complex is the product in which ligands bond with central metal ions by _____.
 [A] H-bonds
 [B] Ionic bonds
 [C] Co-ordinate Covalent bonds
 [D] Covalent bonds
8. What is the hydrogen ion concentration of a solution made by mixing equal valumes of 0.2M hydrochloric acid and 0.4M sodium acetate?
 [A] 6.0×10^{-5}
 [B] 1.8×10^{-5}
 [C] 2.8×10^{-5}
 [D] 6.0×10^{-6}
9. Iodometry titration is that in which _____.
 [A] Direct iodine is used as titrant
 [B] Iodine liberated through chemical reaction is titrated
 [C] Iodine is used as primary standard
 [D] 0.05M iodine solution is used
10. Iodine is standardized with _____ primary standard.
 [A] Sodium thiosulphate
 [B] Arsenic trioxide
 [C] Potassium iodide
 [D] Potassium permanganate
11. For, salt derived from anion of weak acid and cation of weak base ($\text{CH}_3\text{COONH}_4$), If $K_a > K_b$ than solution is _____.
 [A] Neutral
 [B] Acidic
 [C] Buffer
 [D] Basic
12. What is the molarity of an acid made by mixing 25 mL 0.08M acid-1 and 30 mL of 0.11M acid-2?
 [A] 0.083 M
 [B] 0.096 M
 [C] 0.190 M
 [D] 0.099 M
13. How many tablets are generally sampled out from the batch for quantitative analysis?
 [A] 20
 [B] 10
 [C] 15
 [D] 5

14. Which modern concept of acid-base is applicable to non-aqueous acid base titrations?
[A] Bronsted-Lowry theory
[B] Lewis theory
[C] Arrhenius theory
[D] Ionization theory
15. The solubility of difficultly soluble ionogens/precipitates is increased in presence of inert electrolytes, this is called _____.
[A] Diverse ion effect
[B] Common ion effect
[C] Co-precipitation
[D] Digestion
16. In the estimation of iron using redox titration _____ is used as an indicator.
[A] Potassium thiocyanate
[B] Iodine
[C] Starch
[D] KMnO_4
17. Oxidation involves _____.
[A] Loss of electron
[B] Gain of electron
[C] Loss of oxygen
[D] Gain of hydrogen
18. pH condition required in iodine method is _____.
[A] Neutral or slightly acidic
[B] Neutral or slightly basic
[C] Strongly acidic
[D] Strongly basic
19. Solution of acetic acid and sodium acetate is less acidic than solution of acetic acid due to _____.
[A] Common ion effect
[B] Ostwald dilution law
[C] Equilibrium condition
[D] Uncommon ion effect
20. A solution of known concentration is known as _____.
[A] Standard solution
[B] Neutral solution
[C] Buffer solution
[D] Saturated solution

SECTION – II

- Q 2** Attempt any **TWO** of the following
- A** What is acid-base titration? Write a detail note on acid-base indicators. 05
- B** Write a note on the followings. 05
- (i) Hydrolysis of salts.
- (ii) Common ion effect.
- C** What is redox titration? Enlist various types of redox titration. Write a detail note on Iodine titrations. Give applications of iodimetric and iodometric titration. 05
- Q 3** Attempt any **TWO** of the following
- A** 50 mL of 0.1M acetic acid is titrated with 0.1M sodium hydroxide, calculate pH at the various stages of titration and construct the titration/neutralization curve. Suggest suitable indicator for this titration from neutralization curve ($K_a = 1.8 \times 10^{-5}$). 05
- B** What is precipitation titration? Give comparative overview of Mohr's method, Fajan's method and Volhard's method. 05
- C**
- (i) Name the instrument and reagent used for determination of trace amount of water in pharmaceutical products. Outline the quantitative reaction of water with reagent used 05
- (ii) Glacial acetic acid is a leveling solvent for weak bases and differentiating solvent for acidic compounds, justify.

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Define error. Explain types of error and its minimization. 05
- B** Enlist different types of EDTA titrations and discuss any two in detail. 05
- C** What is gravimetric analysis? Enlist various steps and describe each in brief. 05
- D** Discuss applications of (i) complexometric titrations (ii) acid-base titrations. 05
- (i) Give different types of pharmacopoeial water and official quality control tests for waters. 05
- (ii) What do you mean by hardness of water and discuss the methods to remove hardness of water.
- F** What is buffer solution? Derive Henderson-hasselbalch equation for finding pH of buffer solution. 05
- Q 5** Attempt any **FOUR** of the following
- A** Describe various sections of Indian Pharmacopoeia. What are the general contents of monograph? 05
- B** Explain in detail effect of masking and demasking in complexometry. 05
- C**
- (i) Write a note on theory of ionization. 05
- (ii) Explain in brief ionization of weak acids and weak bases. 05
- D**
- (i) Write a note on methods for end point detection in redox titration. 05
- (ii) Give applications of solubility product constant.
- E** Explain the following terms 05
- i) Mean ii) F-test iii) standard deviation iv) Q-test v) t-test.
- F** Write a note on sampling and sampling plans. 05
